

Local Violence and Leader Favorability in Mali

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1 Introduction

Does local political violence relate to how people view the national leader? This report links ACLED violent events, counted by commune and survey wave, to a binary measure of leader favorability from three survey waves in Mali. The main model is a multilevel logistic regression with respondents nested in communes; a design-based regression that uses the strata, PSUs, and weights serves as a check.

2 Data

The analysis file has 5,050 respondents (319 dropped for missing data) in 419 communes. 205 communes were sampled in more than one wave; the rest appear once.

Table 1: Weighted percent favorable by wave, with design-based standard errors (strata, PSUs, weights).

Wave	n	Communes	% favorable	SE	% any event
1	1680	234	53.6	1.7	40.1
2	1695	227	60.9	1.5	38.6
3	1675	226	60.8	1.6	41.3

Exposure. For commune c in wave w , $V_{cw} = \log(1 + E_{cw})$, where E_{cw} is the number of ACLED events. Communes with no ACLED record get $E_{cw} = 0$.

3 Model

Respondent i lives in commune c and was interviewed in wave w . The main model is

$$\text{logit Pr}(y_{icw} = 1) = \beta_0 + \beta_V V_{cw} + \sum_{w=2}^3 \delta_w \text{Wave}_w + \sum_r \rho_r \text{Region}_r + \mathbf{x}_i' \boldsymbol{\beta} + u_c, \quad u_c \sim N(0, \tau^2)$$

- u_c is the **commune random intercept**: communes differ in baseline favorability, and respondents in the same commune are correlated. Because violence is measured at the commune level, this term is what keeps the violence standard error honest.
- Wave and region are **fixed effects**. Region is the design stratum, and eight regions or three waves are too few units to estimate a variance.
- $\mathbf{x}_i$ holds respondent covariates: female, age (per decade), ethnicity, education, urban.

The share of latent variance between communes is $\text{ICC} = \tau^2 / (\tau^2 + \pi^2/3)$. The model is unweighted; clustering is handled by u_c rather than by the PSUs. If the commune is the PSU, the two coincide. If PSUs

are villages within communes, the extra village-level clustering is not modeled, which makes standard errors slightly too small.

Because logit coefficients are hard to read, results are reported as the **average marginal effect** (AME): the average change in the predicted probability of a favorable view per one-unit increase in V_{cw} , with $u_c = 0$.

Within and between communes. Splitting V_{cw} into the commune mean V_c^- and the wave deviation $V_{cw} - V_c^-$ gives separate coefficients β_B and β_W (Bell & Jones, 2015). β_W comes only from communes sampled in more than one wave whose violence changed, and it is not confounded by stable commune characteristics.

Design-based check. The same fixed effects, without u_c , are fit by survey-weighted logistic regression (`survey::svyglm`) with wave-specific strata and PSUs and Taylor-linearized standard errors.

Models were built in steps: M0 null; M1 adds wave and region; M2 adds violence; M3 adds demographics (**main model**); M4 adds local government approval and support for democracy; M5 adds perceived violence and stress. Perceived violence and stress are possible mediators (violence may raise them), so M5 is descriptive. Nested models are compared with likelihood ratio tests. All models are fit by maximum likelihood in `lme4` (Laplace approximation, BOBYQA optimizer).

4 Results

Table 2: Model comparison. LRT compares each model with the previous one. ICC = share of latent variance between communes. AME = change in percentage points of $\Pr(\text{favorable})$ per unit of $\log(1 + \text{events})$, with 95% CI.

Model Added		AIC	BIC	LRT (df)	chi2	p	ICC	Violence	AME [95% CI]
M0	Null	6827.6	6840.7				0.066		
M1	Wave, region	6748.2	6820.0	97.4 (9)		< .001	0.035		
M2	Violence	6683.6	6762.0	66.5 (1)		< .001	0.022	-9.0 [-11.1, -6.9]	
M3	Demographics	6661.6	6811.8	44.0 (11)		< .001	0.021	-9.1 [-11.2, -7.1]	
M4	Local gov, democracy	6371.0	6534.2	294.6 (2)		< .001	0.021	-8.2 [-10.2, -6.1]	
M5	Perceived violence, stress	6350.1	6526.3	24.9 (2)		< .001	0.022	-6.6 [-8.7, -4.5]	

In the main model (M3), each one-unit increase in $\log(1 + E)$, roughly going from no events to two, changes the probability of a favorable view by **-9.1 percentage points** (95% CI -11.2 to -7.1). The commune random-intercept variance is $\hat{\tau}^2 = 0.07$.

Split into its parts, the **within-commune** AME is -8.7 points and the **between-commune** AME is -9.4 points. The design-based model gives -7.5 points (95% CI -9.9 to -5.0). If the design-based and multilevel estimates are close, the conclusion does not depend on how the design is handled.

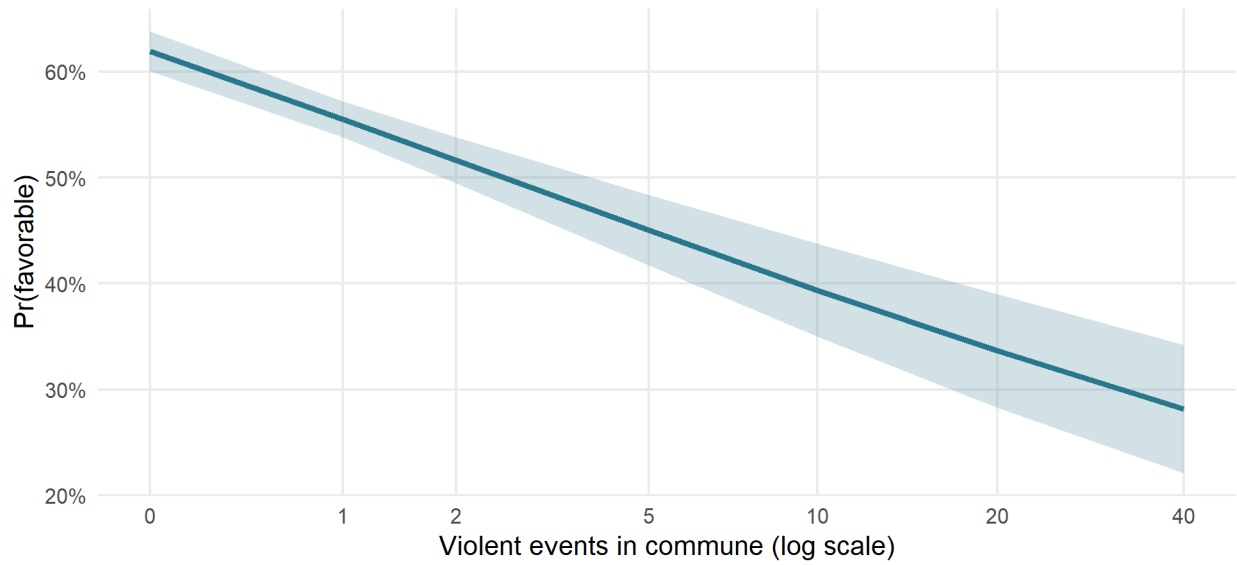


Figure 1: Average predicted probability of a favorable view by commune event count (main model, random intercept at zero, 95% CI).

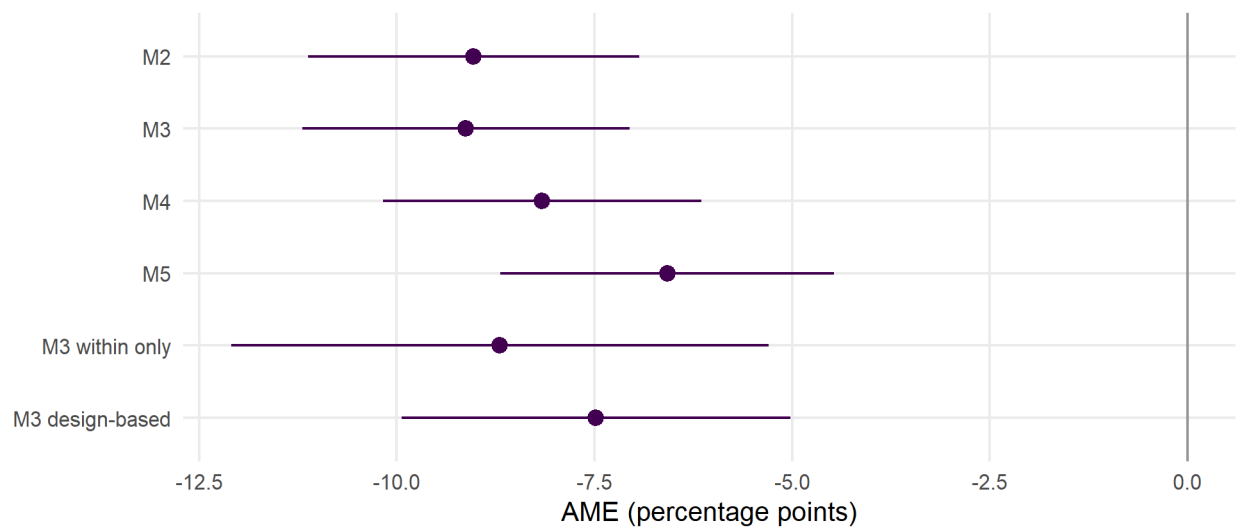


Figure 2: Violence AME (percentage points per unit of $\log(1 + \text{events})$) across models. M4 and M5 add attitudes that violence may itself change, so a smaller estimate there is not evidence against an effect.

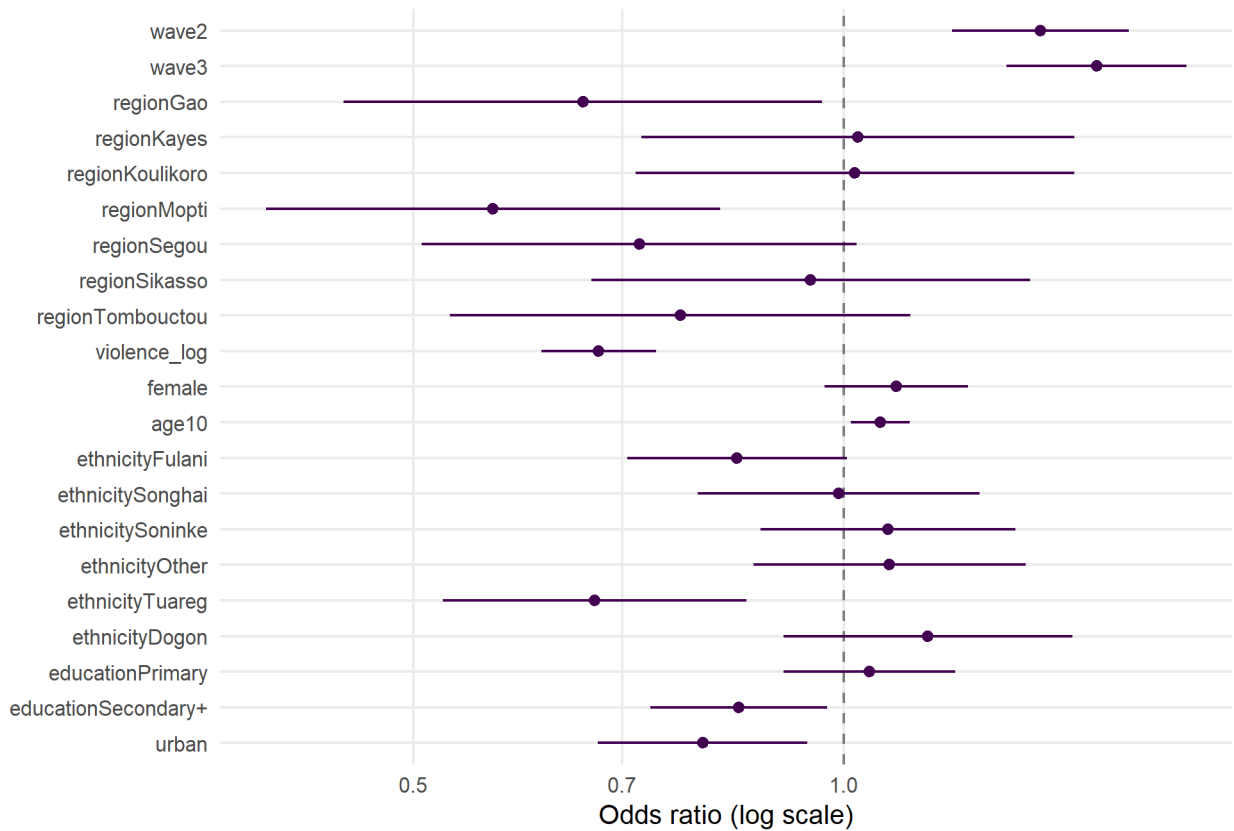


Figure 3: Odds ratios from the main model (M3), 95% CI.

5 Map

Commune, cercle, and region boundaries are from geoBoundaries (Runfola et al., 2020), built into local files by `make_boundaries.R`. The top row shows each commune's weighted percent favorable in each wave (gray: not sampled that wave); the bottom row shows ACLED events on the $\log(1 + E)$ scale, darker red meaning more events. Commune values rest on few respondents, so read the map for broad patterns rather than single communes.

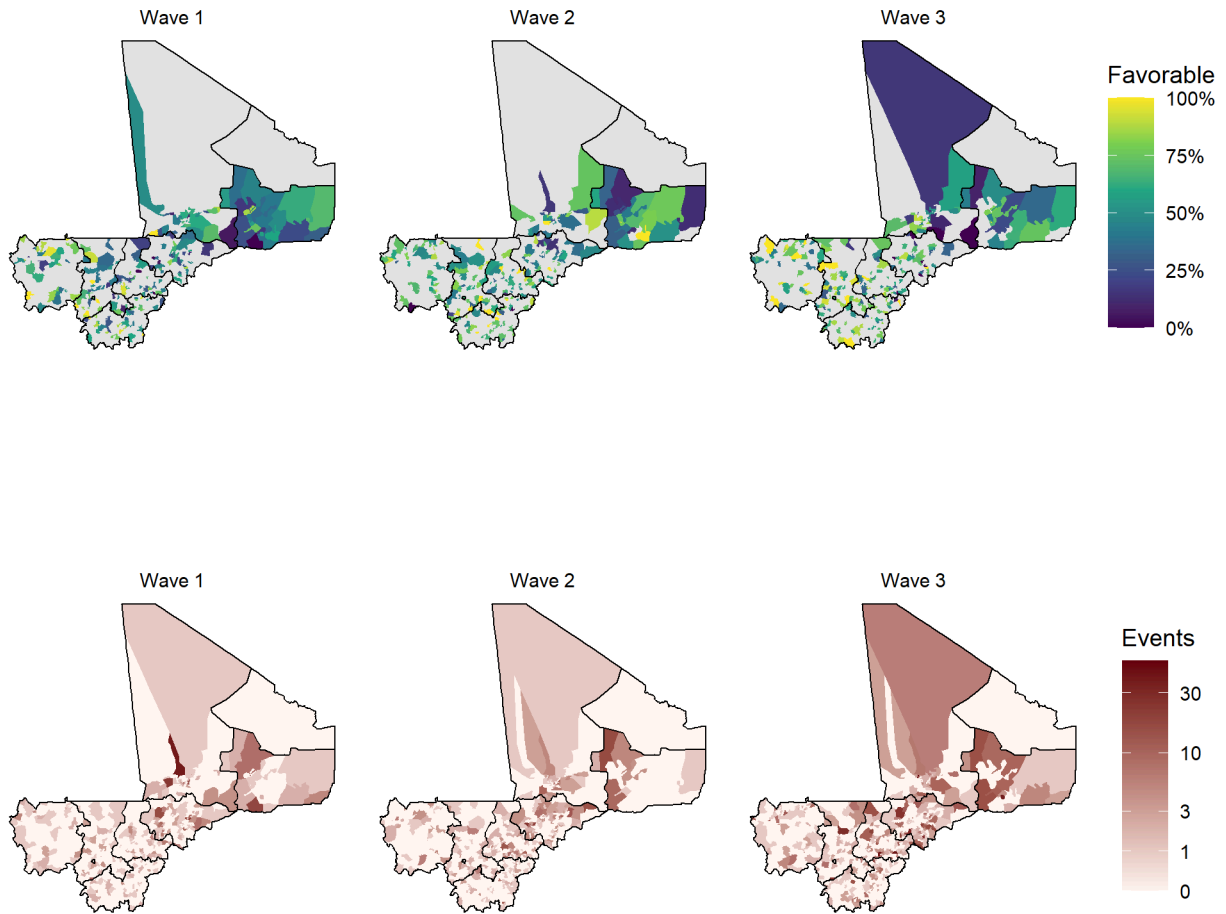


Figure 4: Weighted percent favorable (top) and ACLED events (bottom) by commune and wave. Region boundaries in black.

Table 3: Communes whose region | cercle | commune names did not match the boundary file. Fix spellings using boundaries/mali_admin_dictionary.csv.

Source	Unmatched	Examples
Survey	0	
ACLED	0	

6 Diagnostics

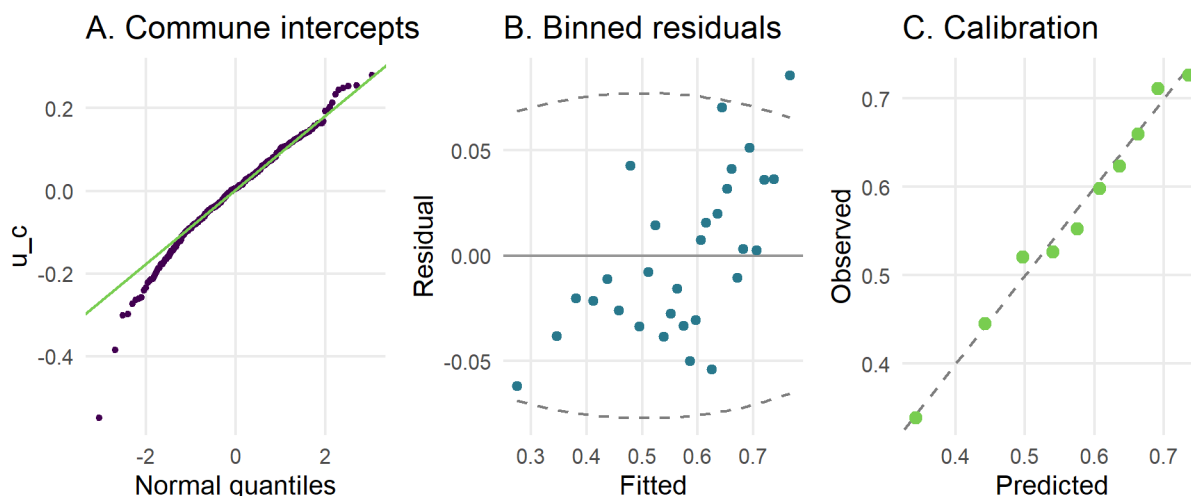


Figure 5: Main-model checks. (A) Commune random intercepts against a normal distribution. (B) Binned residuals: points should fall between the dashed lines with no pattern. (C) Calibration: observed versus predicted by decile.

Convergence: the main model converged without warnings.

7 Limitations

- **Weights.** The multilevel model is unweighted, so it describes associations in the sample. The design-based check shows whether weighting changes the violence estimate.
- **Exposure.** ACLED undercounts events in remote areas, which likely weakens the estimated association. The event window must end before each wave's fieldwork.
- **Coverage.** Insecure areas may be harder to survey, so estimates describe the reachable population.
- **Confounding.** The within-commune estimate controls stable commune differences but not changes over time such as displacement.
- **Mediators.** Attitudes measured alongside the outcome cannot establish mediation.

8 References

- Bell, A., & Jones, K. (2015). Explaining fixed effects: Random effects modeling of time-series cross-sectional and panel data. *Political Science Research and Methods*, 3(1), 133-153.
- Lumley, T. (2010). *Complex surveys: A guide to analysis using R*. Wiley.
- Runfola, D., Anderson, A., Baier, H., Crittenden, M., Dowker, E., Fuhrig, S., Goodman, S., Grimsley, G., Layko, R., Melville, G., Mulder, M., Oberman, R., Panganiban, J., Peck, A., Seitz, L., Shea, S., Slevin, H., Youngerman, R., & Hobbs, L. (2020). geoBoundaries: A global database of political administrative boundaries. *PLoS ONE*, 15(4), e0231866. <https://doi.org/10.1371/journal.pone.0231866>. Mali boundaries (gbOpen, ADM1 to ADM3, source DNCT, DNP, and OCHA Mali; CC BY 4.0): <https://github.com/wmgeolab/geoBoundaries/tree/main/releaseData/gbOpen/MLI>
- Raleigh, C., Linke, A., Hegre, H., & Karlsen, J. (2010). Introducing ACLED: An armed conflict location and event dataset. *Journal of Peace Research*, 47(5), 651-660.